

General Guide for Structuring an M2 Research Report

Purpose of This Note

This is a general reference, not a fixed template. Every project here is different, and this guide is meant to give a sense of shape and expectations, not a form to fill in mechanically. Use whatever parts are useful and skip whatever doesn't fit your topic.

The Usual Shape of a Research Report

Most research reports at this level follow roughly five parts, regardless of subject:

1. **Introduction / Context** - what problem is being addressed, why it matters, and what gap in existing work motivates it.
2. **Background / State of the Art** - what has already been done by others on this or closely related problems, and where the current approaches fall short. Be coherent here, careful what SOTA to mention before the other.
3. **Method** - what was actually designed, built, or tested, described precisely enough that someone else could reproduce it. This is usually the longest and most technical section. along with Methodology or Architecture figure(s).
4. **Results** - what was measured, with concrete numbers, comparisons, and enough detail to judge whether the method worked.
5. **Discussion / Conclusion** - what the results mean, what worked well, what is still open, and what a reasonable next step would be.

This shape holds whether the topic is a machine learning pipeline, a data generation process, a system architecture, or an experimental study. What changes between projects is the content inside each part, not the parts themselves.

A Few General Principles That Tend to Matter

- **Every claim needs a number attached.** “The model performed well” says nothing on its own. “Accuracy improved from X to Y” is something a reader can evaluate.
- **State what was tried and did not work, alongside what did.** A report that only shows successes is less convincing than one that shows the full picture, including dead ends and what was learned from them.
- **Compare against something.** A result only means something in relation to a baseline, an alternative approach, or prior work. A number with nothing to compare it to is hard to judge.

- **Say what remains open.** No project is fully finished at report time. Being explicit about what still needs testing, refining, or extending is expected, not a weakness.
- **One example, to illustrate rather than prescribe:** in a project studying how splitting a large dataset or system across multiple parts affects the quality of some downstream measurement, a report might compare a few different splitting strategies, show which one preserves quality best with numbers, then explain why. That structure, motivation, comparison, explanation, is the general pattern worth reusing, regardless of what the actual subject is.

Length and Tone

There is no fixed page count that applies to every project. What matters more than length is that every section has a clear purpose and every result is backed by evidence. A shorter, well-argued report is generally stronger than a longer one padded with repetition.